Voice Card:

Direct, practical, honest, structured, educational, no buzzwords.

Case Study 1 — FlyRank ML Internship (Current Project)

Problem

FlyRank manages large amounts of SEO content data. Reviewing every page manually is inefficient, so the challenge is to identify which pages should be prioritized for review using historical search performance and machine learning.

What I Did

I completed the onboarding tasks, explored the anonymized SEO dataset, and ran the provided ML pipelines to understand how different models ranked content pages. I analyzed decision trees, compared model performance using Precision@50, experimented with different tree depths, and learned how to avoid data leakage. I also framed a research question focused on content refresh prioritization and began preparing the foundation for my capstone project.

Outcome

I developed a clearer understanding of how machine learning can support business decisions instead of replacing human judgment. I also established a well-defined research direction that will guide the remaining weeks of the internship.

Case Study 2 — Student Dropout Risk Prediction

Problem

Educational institutions often struggle to identify students who are at risk of dropping out before it is too late to intervene.

What I Did

I built a machine learning system using the OULAD dataset to predict student dropout risk. I cleaned and prepared the data, engineered features, trained multiple classification models, compared their performance, and selected XGBoost as the final model. I also developed a Streamlit dashboard so predictions could be explored interactively.

Outcome

The final model achieved strong predictive performance while remaining easy to demonstrate through an interactive interface. The project strengthened my understanding of data preprocessing, model evaluation, and practical machine learning workflows.

Case Study 3 — Indoor Sports Booking Management System

Problem

Manual booking of sports grounds often results in scheduling conflicts, payment issues, and poor user experience.

What I Did

I designed and developed a full-stack web application using ASP.NET Core Web API, React, PostgreSQL, JWT authentication, and Stripe payments. I implemented automatic slot generation, secure authentication, online payments, and a temporary booking hold system to reduce double bookings.

Outcome

The application provides a streamlined booking experience while demonstrating practical backend development, REST APIs, authentication, database design, and payment integration.

Bio

Hi, I'm **Azan Shah**, a Software Engineering student at FAST-NUCES and an ML Intern at FlyRank. I enjoy building software that solves practical problems, whether through backend development with ASP.NET Core or machine learning with Python. I focus on writing maintainable code, learning continuously, and understanding why a solution works rather than simply making it work.

Contact / CTA

I'm currently seeking Software Engineering and Machine Learning internship opportunities where I can contribute to real-world products while continuing to grow as a developer.

Let's connect to discuss software engineering, machine learning, or potential collaboration.

Before / After

Generic AI Version

I am a results-driven software engineer passionate about innovative solutions and leveraging cutting-edge technologies to deliver impactful results.

My Edited Version

I enjoy solving practical software problems by combining backend development with machine learning. I prefer understanding why a solution works, writing maintainable code, and building projects that solve real user needs.